TempTrek

Environmental Monitoring System — Software Specification

Technical specification for version 2.5.5 (Windows x64).

Document type	Software Specification
Application	TempTrek Environmental Monitoring System
Version	2.5.5
Audience	System administrators, QA/validation personnel, and technical support
Platform	Windows 10/11 and supported Windows Server, 64-bit

Scope: This document describes the released application behavior. It is not a replacement for site-specific configuration, risk assessment, IQ/OQ/PQ records, or operating procedures.

1. Purpose and system scope

TempTrek is a local Windows desktop application for acquiring, displaying, recording, alerting on, approving, and reporting environmental measurements from Modbus TCP devices. It is designed for temperature, humidity, power, voltage, frequency, and other mapped sensor channels.

Core functions

- Poll Modbus TCP devices and present current readings on a live dashboard.
- Store readings in daily CSV files at a configured recording interval.
- Provide historical trends, tabular data review, CSV export, and PDF reports.
- Issue configured email and SMS alerts.
- Manage users, permissions, password expiry, approvals, and audit records.
- Create encrypted backups and perform validated, transactional restore operations.

Technology architecture

Layer	Implementation	Responsibility
Desktop interface	Python Tkinter	Login, dashboard, configuration, reporting, and administration screens.
Device communication	pymodbus / Modbus TCP	Background polling of device registers, coils, and discrete inputs.
Data storage	Daily CSV and encrypted JSON configuration	Readings, audit logs, approvals, settings, and reports.
Reporting	ReportLab PDF generation	Trend, data, and audit-trail exports.
Distribution	PyInstaller EXE and Windows MSI	Windows x64 deployment without a user-installed Python runtime.

External interfaces

Interface	Protocol / use
Modbus devices	Modbus TCP; configured host, port, slave ID, function code, register and data type.
Email server	SMTP with verified TLS. Port 465 uses SSL; other supported ports use STARTTLS.
SMS provider	Fast2SMS API when enabled and configured.
Activation service	HTTPS activation validation using the supplied key and the workstation hardware identifier.

2. Functional specification

Monitoring and acquisition

- Each configured device runs in a background worker thread in live or simulation mode.
- Polling is normally once per second; CSV persistence follows the device-specific storage interval.
- Supported functions include FC01 Output Coils, FC02 Input Coils, FC03 Holding Registers, and FC04 Input Registers.
- Supported value types include Boolean, INT16, UINT16, INT32, UINT32, FLOAT32, BCD, and configured scaling/decimal precision.
- A read failure preserves the last good value and visibly marks that channel as stale; it does not create a false zero reading.

Historical data, review, and approval

- Reading files rotate daily and are named by IP address, port, slave ID, and date.
- Users can select devices and exact date/time ranges for review.
- Viewed records retain user, workstation, first/last view, and history information.
- An authorized approver must load and review the exact selection, then re-enter their password before approval.
- Reports containing unapproved readings are identified as DRAFT; fully approved selections are FINAL / APPROVED.
- PDF output is limited to 10 mapped registers and 50,000 records per operation.

Alerts and notification

- Email and SMS rules support enabled state, trigger condition, threshold, cooldown, and recipient configuration.
- Email recipients are validated, de-duplicated, and limited to 50 addresses.
- Scheduled PDF email intervals include 5 minutes, 30 minutes, 1 hour, 6 hours, 12 hours, and 24 hours while the application is running.
- Delivery attempts and failures are recorded in the audit trail and relevant log files.

User access

Users hold explicit permissions for dashboard, device management, historical data, approvals, exports, notifications, user management, audit logs, backups/restores, and company details. The Super Admin is required for privileged user and device-management functions.

3. Data, security, and recovery specification

Storage locations

Location	Content and protection
%APPDATA%\TempTrek\config	DPAPI-protected user accounts, device settings, notification settings, approvals, backup settings, and company settings.
%APPDATA%\TempTrek\live	Daily readings and audit, alarm, email, and SMS log files.
%APPDATA%\TempTrek\reports	Generated reports and retained restore-conflict material.
%ProgramData%\TempTrekBackups	Administrator-owned encrypted portable backup archives.

Security controls

- New account passwords use PBKDF2-HMAC-SHA256 with a random 16-byte salt and 600,000 rounds. Legacy SHA-256 hashes are read only for successful-login migration.
- Configuration is protected using Windows DPAPI, binding it to the Windows user account.
- Configuration writes use a sibling temporary file, flush, and atomic replacement to reduce corruption risk.
- Audit records include timestamp, user, workstation/device data, action, category, prior/updated values, result, details, and a chained integrity hash.
- Sensitive SMTP passwords and SMS API keys are excluded from audit setting snapshots.
- SMTP requires certificate-verified TLS; plaintext fallback is not used.

Backup and restore

- Backup management requires both an elevated Windows administrator session and the application backup permission.
- Portable .ttbackup archives are AES-GCM encrypted with a password-derived key (PBKDF2-SHA256, 600,000 rounds).
- Archive validation rejects unsafe paths, duplicate paths, symbolic links, unsupported members, damaged ZIPs, and excessive archive sizes.
- Restore creates a mandatory safety backup, stages the incoming data, and replaces folders through a rollback journal.
- The merge behavior keeps current configuration authoritative, merges compatible history, preserves conflicts for review, and requires an application restart after restoration.

> **Important limitation:** Windows administrators can change ACLs or delete files. Operational controls and backup retention procedures remain necessary.

4. Deployment and operational constraints

Package and build specification

The Windows x64 release includes a standalone executable and an MSI. The executable requests UAC elevation. Builds are not code-signed. Source builds require Windows Python 3.12 because the MSI builder uses the Windows-only `msilib` module.

```
build.bat
```

The build installs PyInstaller, Pillow, ReportLab, requests, pymodbus, cryptography, and certifi; then builds the EXE and MSI.

Operational constraints

- The application is Windows-specific because local secure configuration relies on Windows DPAPI and backup protection relies on Windows ACLs.
- DPAPI-protected settings must be accessed using the original Windows account. Use an encrypted TempTrek backup for a controlled transfer.
- Automatic backups and scheduled reports run only while TempTrek is running.
- Live device operation, production SMTP delivery, installation/upgrade behavior, and site validation must be verified on the target environment.
- Modbus values and alarm thresholds must be independently checked against the field device before operational use.

Regression coverage

The source tree contains 50 automated unit tests for storage security, backup/restore recovery, password policy, polling safety, monitoring controls, PDF graph output, date formatting, and history merge behavior. Automated tests complement, but do not replace, live hardware, notification, and site validation.

5. Acceptance checklist

1. Verify package checksum and approved release version.
2. Install or launch successfully with expected Windows permissions.
3. Create the Super Admin account and authorized operational accounts.
4. Configure and verify at least one device against a known reference.
5. Confirm records, audit events, notification tests, report generation, and backup creation.
6. Document results under the site's validation and change-control procedure.